

Species-Aware Review: Bacterial Urate Oxidation to Allantoin in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu00230` — Purine metabolism (module area: nucleotide_metabolism) **Review focus:** The uricolysis segment — oxidation of urate (uric acid) to (S)-allantoin **Curation purpose:** Manual module satisfiability and gene-annotation curation

1. Executive Summary

The urate→allantoin (uricolysis) segment of purine catabolism is **only partially satisfiable** in *Pseudomonas putida* KT2440. The organism carries a physically contiguous, well-organized **urate-utilization gene island** spanning `PP_4278–PP_4290` that encodes essentially every component of a functional urate-catabolic system — a uric-acid permease (`PP_4290`), a dedicated GntR-family transcriptional regulator (`PP_4283`), xanthine dehydrogenase and its maturation factor (`xdhAB` / `PP_4278–4280`), and the two committed downstream uricolysis enzymes: **5-hydroxyisourate (HIU) hydrolase** (`pucM` / `PP_4285`, EC 3.5.2.17) and **2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazole (OHCU) decarboxylase** (`pucL` / `PP_4287`, EC 4.1.1.97). Immediately adjacent are the allantoin-degradation enzymes allantoinase (`puuE` / `PP_4286`) and ureidoglycolate lyase (`allA` / `PP_4288`).

The one component that is **missing** is the enzyme that actually commits urate to the pathway: the **entry-step oxidase** that converts uric acid to 5-hydroxyisourate (HIU). A systematic search of the KT2440 proteome found **zero** proteins with a urate-oxidase EC number (EC 1.7.3.x), **zero** proteins carrying the diagnostic Uricase Pfam domain (PF01014 / InterPro IPR002042), and **no** dedicated HpxO-type FAD-dependent urate hydroxylase (EC 1.14.13.113). The FAD-monooxygenase-family proteins that do exist in the genome have established non-urate functions (4-hydroxybenzoate and 6-hydroxynicotinate monooxygenases, ubiquinone-biosynthesis oxidoreductases). The molybdoenzyme `paoABC` (`PP_3308–3310`) that KEGG has

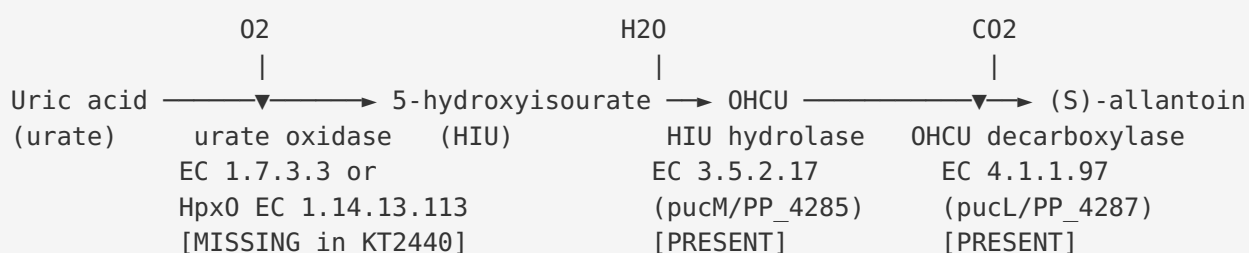
bucketed into purine metabolism is a periplasmic **aldehyde oxidoreductase**, not a urate oxidase. The net picture is a "**pathway hole**" / **genomic fossil**: import and downstream processing machinery are fully intact, but the committed oxidation slot is empty.

For curation, the recommendation is to mark the **entry step (urate → 5-hydroxyisourate)** as **gap** / **candidate_uncertain**, mark the **two downstream steps (HIU → OHCU → allantoin)** as **covered** by **pucM** and **pucL**, flag the **misleading pucL name** (it is a standalone decarboxylase in KT2440, not the bifunctional oxidase+decarboxylase that "pucL" denotes in *Bacillus subtilis*), and flag **paoABC** as **over-bucketed** into uricolysis. Uricolytic growth documented in other *Pseudomonas* species (*P. aeruginosa*, *P. fluorescens*) should be treated as **weak, non-transferable** evidence for KT2440, because the entry-step absence appears to be strain-specific.

2. Target-Organism Pathway Definition

2.1 What the pathway includes

"Bacterial urate oxidation to allantoin" (uricolysis) is the three-enzyme conversion at the heart of aerobic purine-ring catabolism:



The classical entry reaction is catalyzed either by a **cofactor-independent urate oxidase (uricase, EC 1.7.3.3)** carrying the Uricase fold (PF01014), or by the mechanistically distinct **FAD-dependent urate hydroxylase HpxO (EC 1.14.13.113)** found in some enterobacteria. Both make the same unstable product, 5-hydroxyisourate. Historically the spontaneous breakdown of HIU was assumed to yield allantoin directly, but the modern (post-2006) understanding is that two additional, stereospecificity-conferring enzymes — HIU hydrolase and OHCU decarboxylase — complete the conversion to the biologically relevant (S)-(+)-allantoin ([PMID: 16462750](https://pubmed.ncbi.nlm.nih.gov/16462750/)).

2.2 Neighboring pathways to keep separate

- **Purine ring formation upstream of urate** (xanthine dehydrogenase, `xdhAB`) belongs to purine catabolism but is a **separate step** from uricolysis proper. It produces urate; it does not oxidize it.
- **Allantoin degradation downstream** (allantoinase `puuE` /EC 3.5.2.5, ureidoglycolate metabolism, `allA` ureidoglycolate lyase EC 4.3.2.3, `allE` S-ureidoglycine aminohydrolase, urease `ureABC`) is a **distinct module** for nitrogen mobilization from allantoin. It is physically interleaved with the uricolysis genes in KT2440 but should be curated as its own step-set.
- **De novo and salvage purine nucleotide metabolism** (the large `pur*` operon set, `guaAB`, `apt`, `xpt`, `hpt`, nucleotidases, kinases) dominates the `ppu00230` bucket numerically (36 primary genes) but is mechanistically unrelated to urate oxidation. Most of the 65 candidate genes fall here and are **noise** for this specific review.
- **Caffeine / methylxanthine C-8 oxidation** (TmuM, HpxO-like monooxygenases) is a related but separate specialized pathway in other pseudomonads and should not be conflated with core uricolysis.

2.3 Alternate names / database definitions

Uricolysis, urate degradation, purine ring catabolism, "uric acid degradation pathway." KEGG folds all of this into the broad **map00230 (Purine metabolism)** overview, which is why the local bucket `kegg:ppu00230` mixes de-novo synthesis, salvage, and catabolism. The relevant KEGG module for the terminal steps is the uric-acid → allantoin conversion; there was **no module YAML resolved** for this job, so step boundaries had to be reconstructed from the pathway/taxon context.

3. Expected Step Model

Step	Reaction	EC	Canonical gene(s)	Expected in KT2440?
S1 (entry)	Urate + O ₂ → 5-hydroxyisourate	1.7.3.3 (uricase) or 1.14.13.113 (HpxO)	<i>uox</i> / <i>pucL</i> (Bsu) or <i>hpxO</i>	GAP — no encoding gene found
S2	5-hydroxyisourate + H ₂ O → OHCU	3.5.2.17	<i>pucM</i> / <i>uraH</i> / <i>hpxT</i>	Covered — pucM / PP_4285
S3	OHCU → (S)-allantoin + CO ₂	4.1.1.97	<i>pucL(std)</i> / <i>uraD</i> / <i>hpxQ</i>	Covered — pucL / PP_4287
Adjacent up	Xanthine → urate	1.17.1.4	<i>xdhAB</i>	Covered — PP_4278/4279
Adjacent down	Allantoin → allantoate	3.5.2.5	<i>puuE</i> / <i>allantoinase</i>	Covered — PP_4286
Adjacent down	Ureidoglycolate → glyoxylate + urea	4.3.2.3	<i>allA</i>	Covered — PP_4288
Transport	Uric acid import	—	uric-acid permease	Covered — PP_4290
Regulation	Pathway induction	—	GntR-family regulator	Covered — PP_4283

Bottom line: 2 of the 3 core uricolysis steps are encoded; the single committed entry step is absent.

4. Candidate Genes and Evidence

4.1 High-confidence, directly relevant genes

pucM / PP_4285 (Q88F14) — 5-hydroxyisourate hydrolase, EC 3.5.2.17 [S2, COVERED]. UniProt assigns this protein the HIU-hydrolase name and EC directly. It catalyzes the second uricolysis step, hydrolyzing the unstable HIU to OHCU. It sits at the head of the compact PP_4285–PP_4288 uricolysis/allantoin cluster. The two-step "after urate oxidation" model that defines this enzyme's role is established by PMID: 16462750: *"the two proteins... catalyze two consecutive steps following urate oxidation to 5-hydroxyisourate (HIU): hydrolysis of HIU to give... OHCU... and decarboxylation of OHCU to give S-(+)-allantoin."* **Curation note:** high-confidence assignment; the enzyme's substrate (HIU) is only produced if the missing S1 oxidase is supplied.

pucL / PP_4287 (Q88F12) — OHCU decarboxylase, EC 4.1.1.97 [S3, COVERED]. UniProt annotates PP_4287 as the standalone 2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazoline decarboxylase. It catalyzes the third, stereospecificity-determining uricolysis step. **Critical nomenclature caveat:** the gene symbol *pucL* in *Bacillus subtilis* denotes a **bifunctional urate oxidase + OHCU decarboxylase**. In KT2440, PP_4287 is only the **decarboxylase** (a UraD/HpxQ-type enzyme). A curator relying on the *pucL* label could erroneously infer that the entry-step urate oxidase is present. It is not — the oxidase function is not encoded here. This mislabeling is the single most important curation trap in this locus.

xdhA / **xdhB** — PP_4278 (Q88F21) / PP_4279 (Q88F20) — Xanthine dehydrogenase, EC 1.1.7.1.4 [adjacent-upstream, COVERED]. Produces urate from xanthine/hypoxanthine. Present with an accessory maturation factor PP_4280 (Q88F19, XdhC-like). Confirms the pathway is "wired" up to the point of urate production, making the missing oxidase all the more conspicuous.

puuE / PP_4286 (Q88F13) — Allantoinase, EC 3.5.2.5 [downstream, COVERED]. Allantoin → allantoate. Directly adjacent; part of the allantoin-degradation module.

allA / PP_4288 (P59285) — Ureidoglycolate lyase, EC 4.3.2.3 [downstream, COVERED]. Terminal ureidoglycolate cleavage; UniProt has a curated Swiss-Prot entry (P59285), the highest-confidence tier in this locus.

allE / PP_3530 (Q88H35) — S-ureidoglycine aminohydrolase, EC 3.5.3.- [downstream, COVERED but off-island]. Part of allantoin nitrogen mobilization; located outside the main island.

4.2 The urate-utilization island (PP_4278–PP_4290)

The strongest structural evidence in this review is the **synteny** of the locus:

PP_4278	xdhA	– xanthine dehydrogenase subunit	
PP_4279	xdhB	– xanthine dehydrogenase subunit	
PP_4280		– XdhC-like maturation/accessory factor (Q88F19)	
PP_4281	guaD	– guanine deaminase (EC 3.5.4.3)	
PP_4282		– aquaporin Z (Q88F17)	
PP_4283		– GntR-family transcriptional regulator (Q88F16)	← dedicated regulator
PP_4284		– transporter (Q88F15)	
PP_4285	pucM	– HIU hydrolase (EC 3.5.2.17)	← S2 COVERED
PP_4286	puuE	– allantoinase (EC 3.5.2.5)	
PP_4287	puuL	– OHCU decarboxylase (EC 4.1.1.97)	← S3 COVERED
PP_4288	allA	– ureidoglycolate lyase (EC 4.3.2.3)	
PP_4289		– cytochrome c domain protein (Q88F11)	
PP_4290		– uric acid permease (Q88F10)	← import COVERED
		MISSING: urate oxidase / urate hydroxylase (no EC 1.7.3.x, no PF01014, no Hpx0)	← S1 GAP

This is a textbook "**complete island minus one enzyme**" signature: import (permease), regulation (GntR), upstream oxidation (Xdh), and every downstream hydrolytic/decarboxylating/lyase step are present, while the committed urate-oxidation slot is empty.

4.3 Candidate genes that are NOT part of uricolysis (noise / over-bucketing)

The vast majority of the 65 candidate genes belong to **de novo purine biosynthesis** (purF, purD, purN, purT, purL, purM, purK, purE, purC, purB, purH, purA), **GMP/IMP metabolism** (guaA, guaB, gmk), **salvage** (apt, xpt, PP_0747/hpt, PP_0591 adenine deaminase, guaD), **nucleotidases/phosphatases** (yrfG, surE, ushA, PP_2531, yfiH, ppnP, amn, PP_3662), **kinases and NTP housekeeping** (adk, ndk, prs, PP_2744, nrdAB, dgt, mazG, PP_5100, apaH, nudE, nudF, ppx), **signaling** (relA, spoT, cyaA, pde), and **cofactor/sugar metabolism cross-mapped into the bucket** (cysD, cysNC, cpsG, pgm, algC, arcC, ureABC). None of these participate in urate → allantoin oxidation. They are present only because KEGG map00230 is a broad overview map.

paoABC / PP_3308–3310 (Q88HP5/HP4/HP3) — **periplasmic aldehyde oxidoreductase, EC 1.2.99.7 [MIS-BUCKETED]**. These three genes are annotated in the local metadata with primary bucket kegg:ppu00230, and their molybdopterin/2Fe-2S/FAD subunit architecture superficially resembles xanthine-oxidase-family urate-handling enzymes. However, the

characterized *E. coli* ortholog PaoABC is an **aldehyde detoxification** enzyme, not a urate oxidase: [PMID: 27622978](#) describes "*the periplasmic aldehyde oxidoreductase PaoABC from Escherichia coli*" as an exceptional XO-family molybdoenzyme, and [PMID: 24498065](#) characterizes its maturation chaperone PaoD. **paoABC** **does not rescue the entry step** and should not be counted toward uricolysis satisfiability.

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 The entry-step gap (primary finding, F001/F003/F005)

A systematic proteome scan (UniProt SPARQL over UP000000556 / taxon 160488, 2026-08) established the gap on three independent criteria:

1. **Zero proteins with EC 1.7.3.x** (urate oxidase class).
2. **Zero proteins carrying the Uricase Pfam domain PF01014 / InterPro IPR002042.**
3. **No dedicated HpxO-type FAD urate hydroxylase.** Of the five PF01494 (FAD_binding_3 / p-hydroxybenzoate-hydroxylase / HpxO family) proteins in KT2440, four have assigned non-urate functions — Q88H28 (4-hydroxybenzoate-3-monooxygenase), Q88FY2 (6-hydroxynicotinate-3-monooxygenase), Q88CI2 and Q88CI4 (ubiquinone-biosynthesis oxidoreductases) — and the fifth, Q88I03 (generic "Monooxygenase"), is functionally unassigned **but is not clustered with the PP_4285–4290 uricolysis island**, making a cryptic-oxidase hypothesis weak.

The canonical entry reaction requires EITHER a uricase (EC 1.7.3.3) OR an HpxO-type FAD hydroxylase (EC 1.14.13.113) to produce 5-hydroxyisourate before **pucM** and **pucL** can act ([PMID: 16462750](#); [PMID: 18849434](#)). Neither is present. This is a **genuine pathway hole**, not merely an annotation-vocabulary artifact — the search was done at the domain/EC level, which is robust to gene-name inconsistency.

Why HpxO would be easy to miss (and why it still isn't here): [PMID: 18849434](#) notes that in *Klebsiella pneumoniae* the urate-oxidizing HpxO "*did not display any similarity to other reported enzymes known to catalyze these reactions but instead [is] similar to oxygenases acting on aromatic compounds.*" This is exactly why the review checked the aromatic-oxygenase (PF01494) family explicitly — and found only assigned, non-urate members plus one off-island orphan.

5.2 The `pucL` nomenclature trap

As detailed in §4.1, *pucL* in *B. subtilis* = bifunctional urate oxidase + OHCU decarboxylase, but KT2440 PP_4287 is the **standalone decarboxylase only**. Any automated satisfiability check keying on the gene symbol `pucL` would falsely conclude the entry oxidase is present. **This must be corrected in curation.**

5.3 `paoABC` over-bucketing

`paoABC` (PP_3308–3310) is mapped into `ppu00230` but is functionally an aldehyde oxidoreductase (§4.3). Its EC 1.2.99.7 is a promiscuous aromatic-aldehyde dehydrogenase activity. It should be **removed from the uricolysis step model** and re-bucketed to aldehyde metabolism.

5.4 Species-transfer caveat (F004)

Classic physiology (PMID: 4979099) shows *P. aeruginosa* and *P. fluorescens* (and *Klebsiella/Aerobacter*) can grow on uric acid as a nitrogen source, and that this is an **inducible** trait: "*the ability of A. aerogenes, K. pneumoniae, P. aeruginosa, P. fluorescens... to degrade uric acid is an induced property.*" This confirms other pseudomonads have a functional entry oxidase, but it is **weak evidence for KT2440**, where the proteome-level scan finds the oxidase absent. The gap is best interpreted as **strain-specific gene loss** against an intact regulatory/transport/downstream scaffold.

6. Module and GO-Curation Recommendations

6.1 Per-step curation status

Step	Reaction	Recommended status	Rationale
S1 urate → HIU	EC 1.7.3.3 / 1.14.13.113	gap (or candidate_uncertain if the module allows a low-confidence orphan-monooxygenase hypothesis)	No EC 1.7.3.x, no PF01014, no clustered HpxO; the one orphan monooxygenase (Q88I03) is off-island and unassigned
S2 HIU → OHCU	EC 3.5.2.17	covered	pucM /PP_4285, direct UniProt EC assignment, in-island
S3 OHCU → allantoin	EC 4.1.1.97	covered	pucL /PP_4287, direct UniProt EC assignment, in-island (flag name)

6.2 Module-level flags

- **module_needs_revision** — **YES, mildly**. The generic **ppu00230** bucket conflates de-novo synthesis, salvage, and catabolism. For a uricolysis-specific module, restrict to S1–S3 plus explicitly labeled adjacency (Xdh upstream; allantoinase/ureidoglycolate lyase downstream). The 50+ purine-nucleotide-metabolism genes should not be scored against uricolysis satisfiability.
- **Boundaries wrong for this organism?** The KEGG map boundary is generically fine but **too broad** for satisfiability scoring. Recommend a dedicated "urate → (S)-allantoin" mini-module with exactly the three steps above.
- **paoABC re-bucketing**: remove from uricolysis; annotate as aldehyde oxidoreductase.
- **not_expected_in_target_taxon**: Do **not** mark S1 as not-expected. The intact permease + regulator + downstream enzymes argue that urate catabolism *was* expected/functional ancestrally; S1 is a true **gap**, i.e., a fossilized pathway, not a clean absence.

6.3 GO / annotation requests

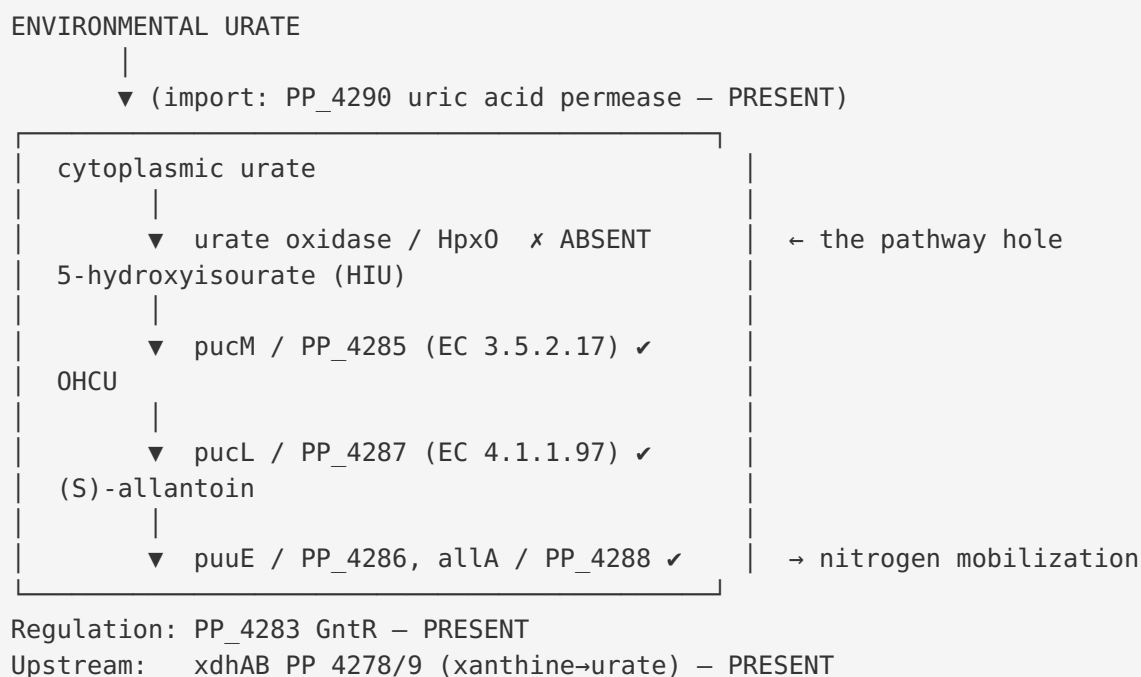
- No new GO terms are required; EC 3.5.2.17 (HIU hydrolase) and EC 4.1.1.97 (OHCU decarboxylase) already exist and map to `pucM` / `pucL`.
- **Request an annotation correction** for PP_4287: retain function "OHCU decarboxylase" but add a caveat that the `pucL` symbol does not imply the bifunctional *B. subtilis* activity.
- **Flag PP_3308–3310** annotations so the `ppu00230` primary-bucket assignment is demoted.

7. Genes to Promote to Full `fetch-gene` Review

Gene / locus	UniProt	Why promote
PP_4287 <code>pucL</code>	Q88F12	Resolve the bifunctional-vs-standalone ambiguity; confirm it is decarboxylase-only and carries no urate-oxidase domain. Highest curation priority.
Q88I03 (orphan "Monooxygenase")	Q88I03	The only functionally unassigned FAD_binding_3 (PF01494) protein; the sole (weak) candidate for a cryptic urate hydroxylase. Determine domain architecture, operon context, and whether any HpxO-like activity is plausible.
PP_4285 <code>pucM</code>	Q88F14	Confirm HIU-hydrolase assignment and in-island role; verify it is not a bifunctional Transthyretin-like protein with additional activity.
PP_3308–3310 <code>paoABC</code>	Q88HP5/ HP4/HP3	Confirm aldehyde-oxidoreductase identity and formally remove from the uricolysis step model.
PP_4290 (uric acid permease)	Q88F10	Confirm substrate specificity; a functioning urate importer with no urate oxidase strengthens the "pathway-hole" interpretation.
PP_4283 (GntR regulator)	Q88F16	Determine regulon; whether it still responds to urate would indicate how recently the oxidase was lost.

8. Mechanistic Model / Interpretation

The KT2440 uricolysis locus is best understood as a **degenerating (fossil) pathway captured mid-loss**:



Every arrow except the first is supported by an encoded, appropriately located enzyme. The strongest interpretation of the intact import + regulation + downstream machinery, combined with the surgically absent oxidase, is that KT2440 **lost the urate oxidase gene specifically** while retaining the rest of the island. This is consistent with the broader theme in the literature that urate-oxidase function is repeatedly and independently lost or degraded across lineages (e.g., the pseudogenization of uricase and the functional decline of downstream enzymes in hominids, [PMID: 41956358](#); and the phylogenetic patchiness of the completed pathway, [PMID: 16462750](#)). Because other pseudomonads retain a functional inducible pathway ([PMID: 4979099](#)), the KT2440 gap is a strain-level rather than genus-level feature.

Practical consequence for module satisfiability: KT2440 is predicted **not to grow on uric acid as a sole source via the classical uricase route** unless a currently unrecognized enzyme (e.g., the orphan monooxygenase Q88I03, or a promiscuous side-activity) supplies HIU. The module should be scored as **incomplete at the entry step**, downstream-complete.

9. Evidence Base

PMID	Title (abbrev.)	How it supports / challenges findings
16462750	<i>Completing the uric acid degradation pathway...</i>	Defines the modern 3-enzyme model (urate oxidase → HIU hydrolase → OHCU decarboxylase); establishes urate oxidase as the canonical entry enzyme whose absence creates the gap, and defines exactly the two steps <code>pucM</code> / <code>pucL</code> catalyze. Core support for F001, F002.
18849434	<i>The hpx system in Klebsiella pneumoniae</i>	Shows the alternative urate oxidizer HpxO is unrelated to classical uricase and resembles aromatic-compound oxygenases — hence it would evade uricase-based annotation. Justifies the explicit PF01494 search. Support for F001, F003.
27622978	<i>E. coli periplasmic aldehyde oxidoreductase PaoABC...</i>	Confirms PaoABC is an aldehyde oxidoreductase (XO-family molybdoenzyme), not a urate oxidase → PP_3308–3310 do not cover the entry step. Support for F003.
24498065	<i>Chaperone PaoD in molybdoenzyme maturation</i>	Reinforces PaoABC identity/maturation as an aldehyde-oxidoreductase system. Support for F003.
4979099	<i>Degradation of uric acid by certain aerobic bacteria</i>	Documents inducible uric-acid degradation in <i>P. aeruginosa</i> , <i>P. fluorescens</i> , <i>Klebsiella</i> , etc. → other pseudomonads are competent, so KT2440's gap is strain-specific. Support for F004.
22609920	<i>Trimethyluric acid monooxygenase TmuM (Pseudomonas CBB1)</i>	HpxO-family flavoprotein monooxygenases oxidize (methyl)urates to (methyl)-HIU; illustrates the enzyme class that would be needed for the entry step but is absent from KT2440. Context for the entry-step gap.
41956358	<i>Structural conservation, functional decline: human OHCU decarboxylase</i>	Illustrates repeated evolutionary loss/decline of uricolytic enzymes after uricase loss — supports the "fossil/pathway-hole" interpretation. Context for F005.
31166064	<i>Ethylene glycol metabolism in P. putida KT2440</i>	Confirms purine/allantoin metabolism is an active, regulated part of KT2440 physiology (GclR/glyoxylate context), consistent with a functional downstream allantoin module. Contextual.

PMID	Title (abbrev.)	How it supports / challenges findings
25678373 , 1158847	Caffeine/N-methylpurine degradation in <i>P. putida</i> strains	Show other <i>P. putida</i> isolates route xanthine through uricase; underscore that uricase presence is strain-dependent within the species. Contextual for species transfer.

10. Limitations and Knowledge Gaps

- 1. Homology/domain-based negative evidence.** The entry-step gap rests on absence of EC 1.7.3.x, PF01014, and a clustered HpxO — i.e., a *negative* genomic result. It is strong but cannot fully exclude a highly divergent, unrecognized urate oxidase (e.g., a promiscuous activity of the orphan monooxygenase Q88I03). Negative homology evidence is inherently weaker than a positive biochemical assay.
- 2. No experimental growth data for KT2440 on urate.** The prediction that KT2440 cannot use uric acid as an N source via the classical route is inferred, not measured here.
- 3. No module YAML was available,** so the step model was reconstructed from KEGG/pathway context rather than a curated module definition; exact step boundaries are review-defined.
- 4. Regulator/permease function inferred from annotation,** not from transcriptomics — the "intact but idle" island interpretation would be strengthened by expression data.
- 5. Species-transfer uncertainty.** Other *Pseudomonas* competence for urate is documented, but strain-to-strain gene content in *P. putida* varies; KT2440-specific conclusions should not be generalized to all *P. putida*.

11. Proposed Follow-up Experiments / Actions

Curation actions (immediate): - Mark uricolysis **S1** = **gap**, **S2/S3** = **covered** in the module. - Add nomenclature caveat to PP_4287 (**pucL** ≠ bifunctional oxidase). - Demote **paoABC** (PP_3308–3310) from the **ppu00230**/uricolysis bucket. - Promote PP_4287, Q88I03, PP_4285, PP_4290, PP_4283 to full **fetch-gene** review (§7).

Bioinformatic follow-up: - Structural modeling (AlphaFold/Foldseek) of the orphan monooxygenase **Q88I03** against HpxO/TmuM to test any cryptic urate-hydroxylase potential. - Comparative genomics across *P. putida* strains and close relatives to date the oxidase loss and check for a pseudogene remnant in the PP_4278–4290 interval. - Verify the uric-acid permease (PP_4290) substrate assignment and the GntR (PP_4283) regulon by operon/motif analysis.

Experimental (definitive): - **Growth assay:** test KT2440 on uric acid as sole N (and C) source, $\pm$ induction, to directly confirm the entry-step gap phenotype. - **Heterologous complementation:** express a known urate oxidase (e.g., *K. pneumoniae* HpxO or a bacterial uricase) in KT2440 and test for restored urate growth — would confirm that S2/S3 and transport are functionally competent and only S1 is missing. - **Biochemical assay** of recombinant Q88I03 for urate-hydroxylase activity to close the "cryptic oxidase" possibility.

12. Key References

- [PMID: 16462750](#) — Completing the uric acid degradation pathway through phylogenetic comparison of whole genomes.
- [PMID: 18849434](#) — The *hpx* genetic system for hypoxanthine assimilation in *Klebsiella pneumoniae*.
- [PMID: 27622978](#) — The *E. coli* periplasmic aldehyde oxidoreductase (PaoABC), XO-family molybdoenzyme.
- [PMID: 24498065](#) — Molecular chaperone PaoD in molybdoenzyme maturation.
- [PMID: 4979099](#) — Degradation of uric acid by certain aerobic bacteria (inducible in *Pseudomonas*).
- [PMID: 22609920](#) — Trimethyluric acid monooxygenase (HpxO-family) in *Pseudomonas* sp. CBB1.
- [PMID: 41956358](#) — Structural conservation, functional decline of human OHCU decarboxylase.
- [PMID: 31166064](#) — Ethylene glycol / purine-allantoin metabolism in *P. putida* KT2440.

Prepared for manual module satisfiability and gene-annotation curation. Evidence for the entry-step gap is proteome-level (UniProt SPARQL over UP000000556 / taxon 160488) and domain-based; downstream-step coverage is from direct UniProt EC assignments; species-transfer claims from other Pseudomonas are flagged as weak.

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